INCIDENT RESPONSE REPORT

IR-001 — SSH Brute Force Investigation

Wazuh • Ubuntu Linux • SSH Authentication

Incident ID	IR-001
Status	Investigating
Severity	High (provisional)
Affected asset	ubuntu_vm2 / 192.168.34.143
Source IP	192.168.34.138
Date	17 October 2026
Analyst	Vipin Kumar

1. Executive Summary

Wazuh detected repeated SSH password failures against Ubuntu endpoint ubuntu_vm2 (192.168.34.143) from source 192.168.34.138. Custom rule 100200 generated a Level 10 alert. Endpoint logs also show a successful password-based SSH login for account vm2 from the same source at 12:11:35.546 (UTC+05:30), followed by a sudo session at 12:12:09.

The sequence is suspicious and requires validation. The evidence does not establish whether the source was authorized or whether the account was compromised. Verdict: Suspicious activity — investigation required.

2. Incident Details

Field	Value
Incident ID	IR-001
Incident type	SSH authentication attack investigation
Severity	High (provisional; validate authorization)
Status	Investigating
Detection platform	Wazuh
Wazuh manager	vm1
Affected asset	ubuntu_vm2 — 192.168.34.143
Source IP	192.168.34.138
Target account	vm2
Log source	/var/log/auth.log
Custom detection	Rule 100200 (Level 10)
Related rule	100201 (successful login after brute-force alert; validation incomplete)
Investigation date	17 October 2026 (UTC+05:30)
Analyst	Vipin Kumar

3. Detection Summary

Rule 100200 is configured to correlate SSH authentication failures from the same source: frequency 5, timeframe 180 seconds, ignore 180 seconds, parent rule 5760, and same-source-IP correlation. Wazuh shows five visible failed-password records in the alert context and the alert was generated at Level 10.

The configuration and correlation behavior should still be tested with wazuh-logtest and controlled events. The existence of an alert alone does not validate every threshold or suppression condition.

4. Key Evidence

Time (UTC+05:30)	Observed event
11:56:00.762	Failed SSH password for vm2 from 192.168.34.138
11:56:02.964	Failed SSH password for vm2 from 192.168.34.138
11:56:07.020	Failed SSH password for vm2 from 192.168.34.138
11:56:08.662	Failed SSH password for vm2 from 192.168.34.138
11:56:10.848	Failed SSH password; associated with Wazuh rule 100200
12:10:51–12:11:07	Further failed SSH password attempts from 192.168.34.138
12:11:35.546	Successful password-based SSH login for vm2 from 192.168.34.138, port 33084
12:11:35.552	SSH session opened for vm2
12:12:09.442	Sudo session opened by vm2 to run tail -n 50 /var/log/auth.log

The source IP, target account, and activity are observed facts. They are not independently confirmed malicious indicators.

5. Findings

Finding 1 — Repeated SSH authentication failures

Multiple failed password events were recorded during two periods: approximately 11:55:58–11:56:15 and 12:10:51–12:11:08. The source was 192.168.34.138 and the target account was vm2. Confidence that repeated failures occurred: high; intent remains unconfirmed.

Finding 2 — Wazuh alert generated

Wazuh generated custom rule 100200, Level 10, with description “Multiple SSH authentication failures from same source - possible brute force.”

Finding 3 — Successful login after failures

The endpoint logged “Accepted password for vm2 from 192.168.34.138 port 33084 ssh2” at 12:11:35.546. Authentication success is confirmed; authorization and compromise are not.

Finding 4 — Sudo activity after login

At 12:12:09, a sudo session was opened by vm2 to run /usr/bin/tail -n 50 /var/log/auth.log. This can be legitimate troubleshooting or investigation; context is required.

6. MITRE ATT&CK

Technique	Assessment
T1110 — Brute Force (Credential Access)	Repeated password failures are consistent with brute-force behavior; automated or malicious intent is not proven.
T1078 — Valid Accounts	Successful authentication occurred, but unauthorized use of valid credentials is not established. Treat as a hypothesis until validated.

7. Impact Assessment

Confirmed: repeated SSH authentication failures, one Wazuh Level 10 alert, a successful SSH login, and a subsequent sudo session.

Not established: unauthorized access, account compromise, persistence, lateral movement, data exfiltration, or malicious commands. No malicious file hash, domain, or malware artifact was identified in the supplied evidence.

8. Analyst Verdict

SUSPICIOUS ACTIVITY — INVESTIGATION REQUIRED.

The successful login after repeated failures increases concern, but the available evidence cannot distinguish an authorized lab test or administrative action from unauthorized access. Keep the case open until the source machine and login are confirmed.

9. Recommended Response

1. Confirm whether 192.168.34.138 is the authorized lab/Kali or administrator machine and whether the successful login was expected.
2. Preserve SSH/authentication logs, sudo records, and relevant Wazuh alerts.
3. Review commands, processes, sessions, account changes, SSH keys, cron jobs, services, and subsequent logins around and after 12:11:35.

4. If the login is unauthorized or cannot be explained, restrict source access, terminate suspicious sessions where appropriate, reset credentials from a trusted session, and check for persistence.
5. Do not block the source or declare compromise solely because this alert fired; document the authorization check and response actions.

10. Detection Validation & Improvements

6. Run wazuh-logtest against representative failed SSH events and verify rule 100200 matching and source-IP correlation.
7. Test the five-event/180-second threshold and 180-second suppression behavior using controlled lab events.
8. Test rule 100201 with a successful login following a qualifying alert within its configured five-minute window. The supplied screenshots show three rule 100201 alerts in the selected time range, but the matching event details and correlation should be validated before treating this as proven causal correlation.
9. Correlate events by source IP, username, host, and event time; distinguish endpoint event timestamps from Wazuh indexed timestamps.
10. Add source asset identity and expected-login context to triage notes where possible.

11. Conclusion

The supplied evidence confirms repeated SSH authentication failures against ubuntu_vm2 from 192.168.34.138, a Wazuh Level 10 alert under rule 100200, a subsequent successful password-based SSH login for vm2, and a sudo session shortly afterward. This is suspicious and merits follow-up, but compromise is not confirmed. Close the incident only after validating whether the login was authorized and reviewing post-login activity.

12. Evidence Index

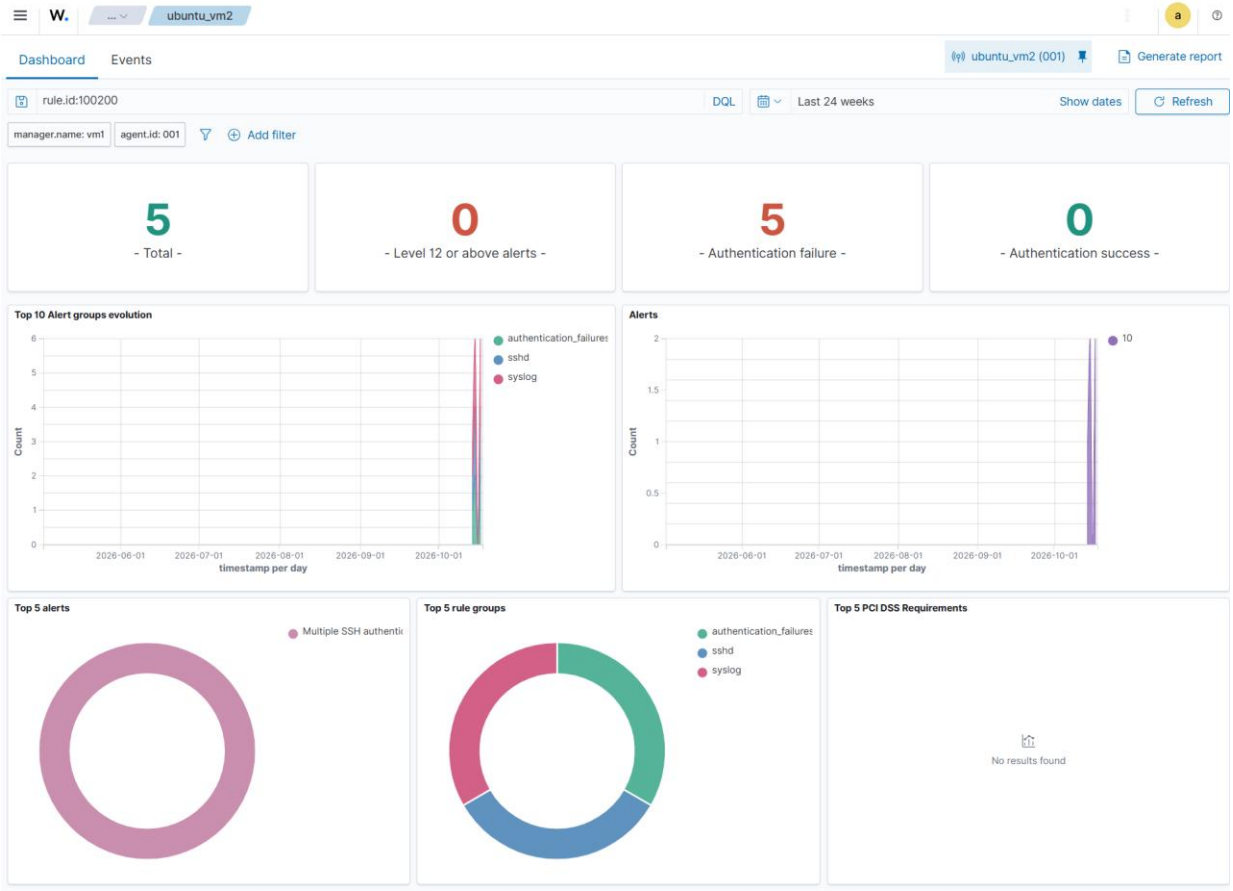
Evidence ID	Filename	Description
E01	E01-wazuh-alert.png	Wazuh alert results for rule 100200
E02	E02-alert-details.png	Wazuh document details including source/target metadata
E03	E03-alert-search-results.png	Search results showing five rule 100200 alerts
E04	E04-successful-login-correlation.png	Search results for rule 100201 showing three alerts
E05	E05-custom-detection-rule.png	Screenshot of custom rule configuration
E06	E06-wazuh-dashboard.png	Wazuh dashboard view filtered to rule 100200

Evidence note: these are the screenshots supplied for this report. The supplied set does not include a standalone full terminal capture of all authentication and sudo log lines or a wazuh-logtest validation transcript. Add those only after capturing the actual output.

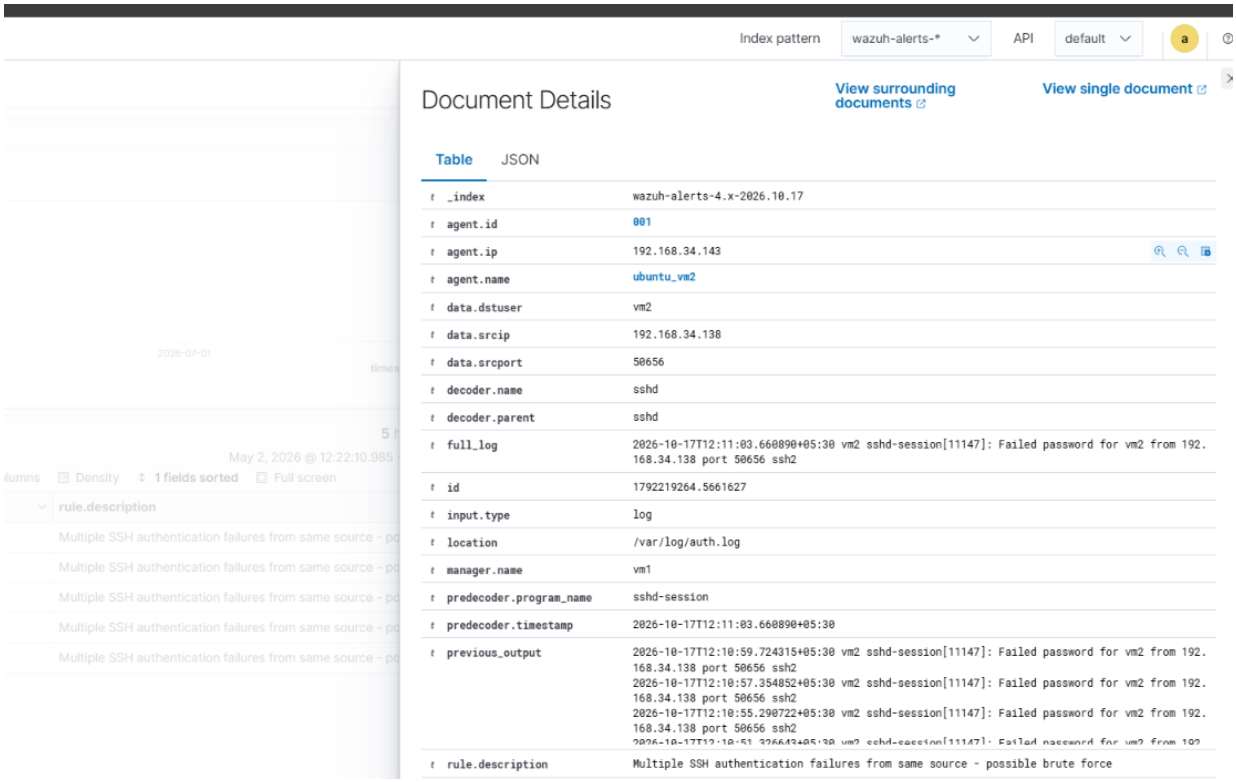
Appendix A — Screenshot Evidence

Screenshots are included in the evidence/ folder. Each image is presented as supplied; captions describe what is visible and do not imply uncollected evidence.

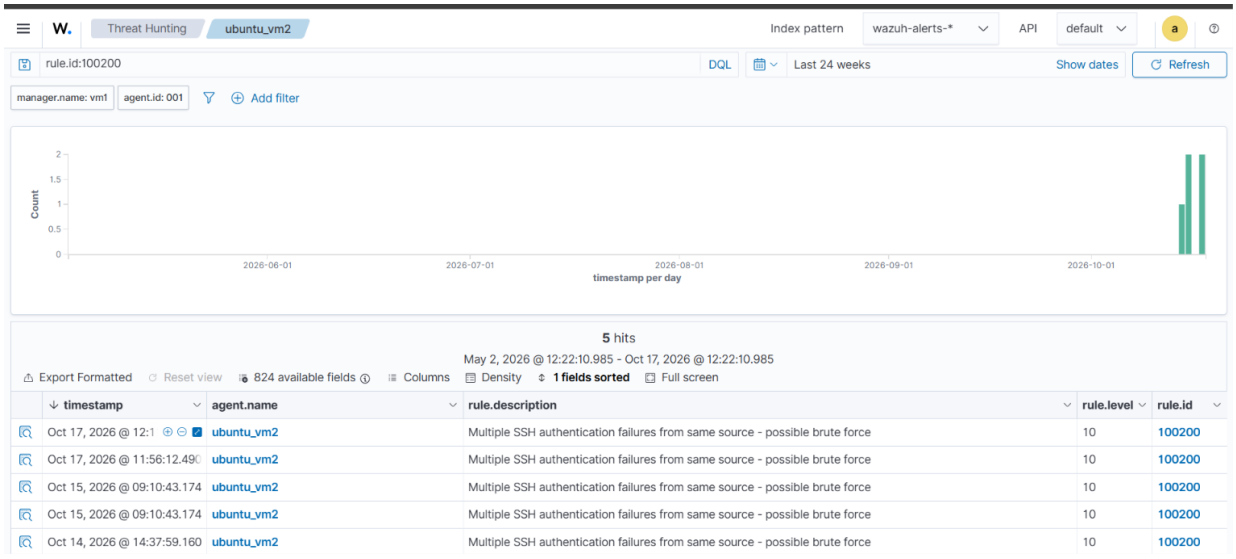
E01-wazuh-alert-dashboard.png



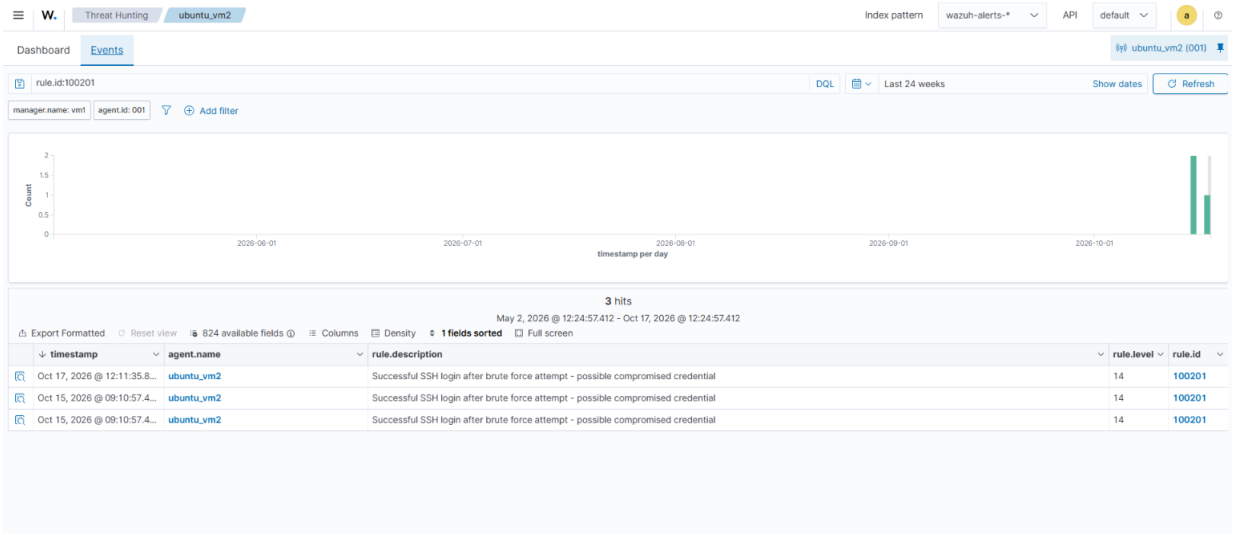
E02-alert-details.png



E03-alert-search-results.png



E04-successful-login-correlation.png



E05-custom-detection-rule.png

```
<!-- custom ssh brute force -->
<group name="syslog,sshd,">
  <rule id="100200" level="10" frequency="5" timeframe="180" ignore="180">
    <if_matched_sid>5760</if_matched_sid>
    <same_source_ip />
    <description>Multiple SSH authentication failures from same source - possible brute force</description>
    <mitre>
      <id>T1110</id>
    </mitre>
    <group>authentication_failures,</group>
  </rule>

  <!-- succssful ssh after multiple failed login -->
      <rule id="100201" level="14" timeframe="300">
    <if_sid>5715</if_sid>
    <if_matched_sid>100200</if_matched_sid>
    <same_source_ip />
    <description>Successful SSH login after brute force attempt - possible compromised credential</description>
    <mitre>
      <id>T1110</id>
      <id>T1078</id>
    </mitre>
    <group>authentication_success,brute_force_success,</group>
  </rule>
</group>
```